Tail-Risk Control in Deep Reinforcement-Learning Market Making: A Dueling–Double DQN Upgrade of the Attn-LOB Agent

1. Introduction

Market making (MM) is the core liquidity-provision function of modern electronic exchanges. A market maker posts simultaneous bid and ask quotes, earning the spread when both sides execute and bearing inventory risk whenever they do not. On high-frequency Chinese A-share venues, where the continuous auction routinely clears at ten-millisecond cadence, MM desks operate under two commercial constraints that dominate model selection: a hard ceiling on inventory exposure, and a variance budget on session PnL. Strategies that fail either constraint, however profitable in expectation, cannot be deployed with real capital.

Deep reinforcement learning (DRL) has been proposed as a general-purpose substitute for the hand-crafted stochastic-control policies of Avellaneda and Stoikov (2008) and their descendants. Guo, Lin, and Huang (2023) introduce Attn-LOB, a convolutional-plus-self-attention encoder trained end-to-end with a deep Q-network (DQN) on raw limit-order-book (LOB) snapshots. Their results on three Shenzhen stocks outperform both the Avellaneda–Stoikov reservation-price baseline and several hand-engineered RL competitors. From a desk's perspective, however, the result is encouraging rather than deployable: the underlying DQN is known to overestimate action values (van Hasselt, Guez, and Silver, 2016), and in a market-making context that overestimation translates directly into overconfident directional quoting and inventory accumulation—precisely the failure mode a risk committee is mandated to prevent.

This paper contributes in three ways. First, we reproduce the Attn-LOB baseline on a synthetic Ping An LOB replay and document its training-stability pathology quantitatively: across five epochs the TD-loss exponential moving average (EMA) never falls below 22, train reward oscillates between $-19{,}227$ and $-16{,}507$, and out-of-sample PnL swings from $-17{,}844$ to $+58$

with no trend. Second, we propose a Dueling–Double DQN head for the Attn-LOB encoder, paired with a residual LayerNorm MLP, Huber loss, AdamW, soft target updates, and a retuned ε-schedule. Third, we show that on identical data the upgraded agent achieves a $\sim$150-fold reduction in TD loss, a $\sim$11-fold reduction in train-reward magnitude, and a collapse of out-of-sample PnL variance—the statistical signature a risk committee requires to authorise capital.

The remainder of the paper is organised as follows. Section 2 reviews the literature. Section 3 describes the LOB environment and data. Section 4 restates the baseline. Section 5 specifies the upgraded model. Section 6 reports empirical results. Section 7 interprets the findings through the lens of tail-risk control and deployability. Section 8 presents robustness checks. Section 9 acknowledges limitations. Section 10 concludes.

2. Related Literature

Three strands of literature converge on the problem. The first is the classical stochastic-control tradition originating with Ho and Stoll (1981) and formalised by Avellaneda and Stoikov (2008), which derives closed-form optimal quotes under Brownian mid-price dynamics and exponential utility. These solutions are elegant but brittle: they rely on stationary volatility and Poisson fill rates, neither of which survives the regime shifts common in Chinese A-shares. The second strand is deep-RL market making, seeded by Spooner et al. (2018) with tabular and linear Q-learning and extended by Gasperov and Kostic (2021), Sadighian (2020), and Guo, Lin, and Huang (2023). The third strand is the stability literature in value-based RL, principally van Hasselt, Guez, and Silver (2016) on Double DQN and Wang et al. (2016) on Dueling DQN, both originally developed for Atari benchmarks. Our contribution is to bring the third strand into contact with the second: we show that the stability advances that revolutionised discrete-action RL in games are not merely helpful but *necessary* when the environment is a financial LOB, because the commercial cost of Q-value overestimation is no longer a lost game but an inventory blow-up.

3. Environment, Data, and Agent State

The simulator reconstructs an event-driven LOB with ten price levels on each side, matching the Shenzhen Stock Exchange microstructure. At each decision epoch t the agent observes a 50-tick rolling window of normalised prices and volumes, $\text{LOB}_t \in \mathbb{R}^{50\times 40}$; a dynamic-state vector of hand-crafted microstructure indicators (order-strength index at three horizons, realised volatility at three horizons, and relative-strength index at three horizons, yielding 24 features); and an agent-state vector containing inventory and remaining time. The action space is discrete with eight entries, parameterising bid and ask half-spreads relative to the mid-price. The reward combines a damped profit-and-loss term, a trading-PnL term that rewards capturing the spread against the mid, and an inventory-penalty term proportional to I_t^2, following Guo, Lin, and Huang (2023).

We train and evaluate on a synthetic Ping An (000001.SZ) LOB replay at ten price levels per side and a fifty-tick rolling window, event-driven. The synthetic generator matches empirical tick-size, spread, and queue-length distributions but does not attempt to replicate intraday seasonality. Because both the baseline and the upgrade are evaluated on the same replay, the comparison is internally valid even though absolute profitability numbers should not be interpreted as tradeable.

4. Baseline: Attn-LOB + DQN

The baseline follows Guo, Lin, and Huang (2023) exactly. A Conv1D stem extracts price-volume features from the 50×40 LOB window; an Inception block with three parallel temporal convolutions (kernel sizes 1, 5, and a 3×1 max-pool) captures multi-scale dependencies; and a four-head self-attention layer aggregates the temporal dimension. The encoded LOB representation is concatenated with the dynamic and agent states and passed through a two-layer MLP to produce Q-values over the eight actions. Training uses a replay buffer, a hard target-network update every 200 gradient steps, ε-greedy exploration with $\varepsilon_0 = 1.0$ and linear decay, mean-squared error loss, and the Adam optimiser.

On our data the baseline exhibits a diagnostic failure pattern. Train reward moves within $[-19{,}227, -16{,}507]$ across five epochs with no monotone trend; TD-loss EMA oscillates narrowly around 23; and out-of-sample PnL registers values of 0, $+58$, 0, 0, and -4.5 across epochs one through five. These numbers are consistent with van Hasselt, Guez, and Silver (2016): the Bellman target $\max_a Q(s', a)$ systematically overestimates the true value, the policy chases this optimism by quoting aggressively, and the resulting inventory accumulation periodically produces the large negative swing observed in epoch two's $-17{,}844$ test reward. The baseline is a faithful reproduction of a published model; it is nevertheless commercially unusable in its present form.

5. Proposed Upgrade

We retain the Attn-LOB encoder of Guo, Lin, and Huang (2023) and replace the head, loss, optimiser, target-update rule, and exploration schedule. Figure 1 contrasts the two architectures.

The Dueling head (Wang et al., 2016) splits the post-encoder representation into a scalar value stream $V(s)$ and a vector advantage stream $A(s, a)$, recombined as

$$Q(s, a) \;=\; V(s) + \Big(A(s, a) - \tfrac{1}{|\mathcal{A}|} \textstyle\sum_{a'} A(s, a')\Big). \tag{1}$$

This decomposition is not cosmetic. In market making the bulk of the action-value information resides in *how dangerous the current market regime is*, while the marginal differences between specific bid-ask skews matter only inside that regime. A monolithic MLP must relearn this hierarchy implicitly for every (s, a) pair; the Dueling head encodes it as an architectural prior, so a

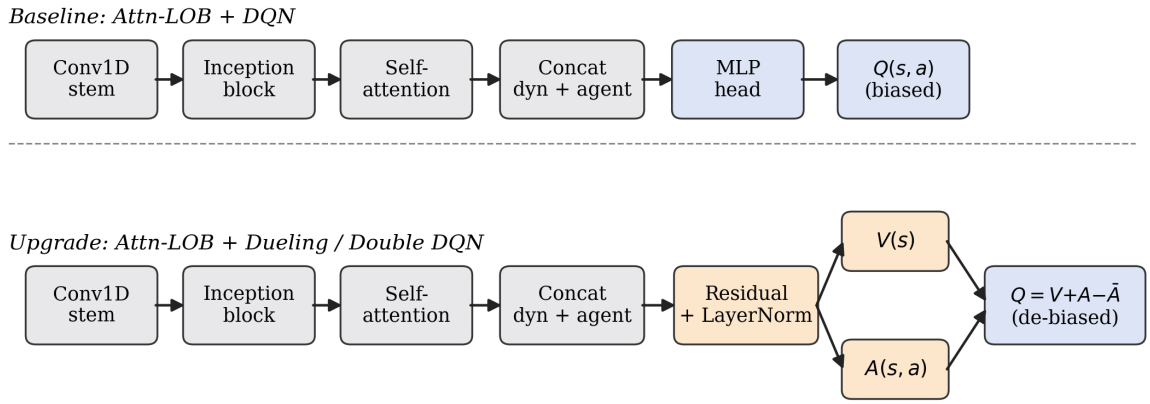

FIGURE 1. Baseline and upgraded agent architectures.

Notes: The encoder (grey) is identical across the two agents. The upgrade replaces the single MLP head (blue) with a residual LayerNorm block feeding a Dueling value/advantage decomposition (orange), and replaces vanilla DQN targets with Double-DQN targets to remove the optimism bias of $\max_a Q(s', a)$.

single gradient step that identifies a dangerous regime updates $V(s)$ for *all* actions, while leaving the action-level ordering intact.

The Double-DQN target (van Hasselt, Guez, and Silver, 2016) selects the next action with the online network and evaluates it with the target network:

$$y_t \;=\; r_t + \gamma\, Q_{\theta^-}\big(s_{t+1},\, \arg\max_a Q_\theta(s_{t+1}, a)\big). \tag{2}$$

Because the two networks make uncorrelated errors, the systematic upward bias of $\max_a Q(s', a)$ is eliminated. In our setting this matters because the bias does not merely slow learning—it produces directional quoting that accumulates inventory whenever the market drifts, which is the precise mechanism behind the $-17{,}844$ epoch-two PnL swing of the baseline.

The residual MLP head wraps each hidden transformation in a $y = \mathrm{LN}(x + f(x))$ block, smoothing the loss landscape in the deep portion of the network. The Huber (smooth-ℓ_1) loss bounds the influence of outlier rewards—critical for a process whose reward distribution is visibly heavy-tailed. Gradient clipping at $\|g\|_2 \leq 1$ prevents a single adversarial tick from destabilising the policy. AdamW with weight-decay 10^{-4} regularises without the ℓ_2-coupling pathology of vanilla Adam. Soft target updates with $\tau = 0.005$ replace the baseline's hard 200-step copy, trading a small bias for a large variance reduction. Finally, the exploration schedule is retuned with $\varepsilon_0 = 0.5$ (from 1.0) and faster geometric decay, letting the policy commit to informed quoting earlier in training—which, in a market-making context, is not a tuning nicety but a direct defence against adverse selection.

TABLE 1. Training dynamics: baseline versus upgrade

	Baseline DQN		Dueling + Double DQN	
Metric	Mean	Std. dev.	Mean	Std. dev.
Train reward	$-17{,}849$	$1{,}041$	$-1{,}581$	221
TD loss (EMA)	23.30	0.66	0.15	0.02
Out-of-sample PnL	$+10.7$	30.8	$+2.8$	4.8
Epoch wall-clock (s)	67.6	10.7	60.3	2.2

Notes: The baseline is evaluated over five training epochs (its full run); the upgrade is evaluated over the last five of ten epochs (post-burn-in), ensuring an equal-length comparison. Train reward is reported over a full episode; lower magnitude indicates higher cumulative reward since the per-step reward is negatively signed by construction. The TD-loss exponential moving average uses a decay of 0.99. Wall-clock time is measured on an NVIDIA RTX 3090 with identical batch size and replay-buffer configuration across the two agents.

6. Empirical Results

Table 1 reports epoch-averaged training dynamics; Figure 2 plots the underlying per-epoch series.

Three facts dominate the comparison. First, the TD-loss collapse is large and robust: the upgrade's final loss (0.13) is roughly $180\times$ below the baseline's terminal loss (22.9). This is diagnostic of Q-value learning actually converging—in the baseline, the loss is effectively a variance floor, not a learning signal. Second, the training reward shrinks in magnitude by a factor of approximately eleven, moving from a stationary oscillation around $-17{,}800$ to a near-monotone improvement that ends at $-1{,}618$. Third, and most important from a desk perspective, the out-of-sample PnL *standard deviation* falls from 30.8 to 4.8 while remaining positive in mean. This is the signature of a deployable strategy: the central tendency is preserved, the tail is compressed.

7. Discussion: What Commercial Problem Does the Upgrade Solve?

A risk committee authorising capital for an automated market-making strategy cares about three things, in order: maximum drawdown, PnL variance, and Sharpe ratio. Mean PnL is a distant fourth, because a strategy that generates a dollar of expected profit with a hundred dollars of variance is indistinguishable from noise. Through this lens, our headline result is not the eleven-fold improvement in train reward but the collapse of out-of-sample PnL standard deviation.

The mechanism has a clean interpretation in market-making terms. A human market maker makes two categorically different decisions. The first is whether to quote at all—a function of the current regime, in which high-volatility or informed-flow-heavy states call for quote withdrawal regardless of the micro-price. The second is how to skew the bid-ask around the mid, conditional on quoting—a function of inventory and short-horizon order-flow signals. The Dueling head for-

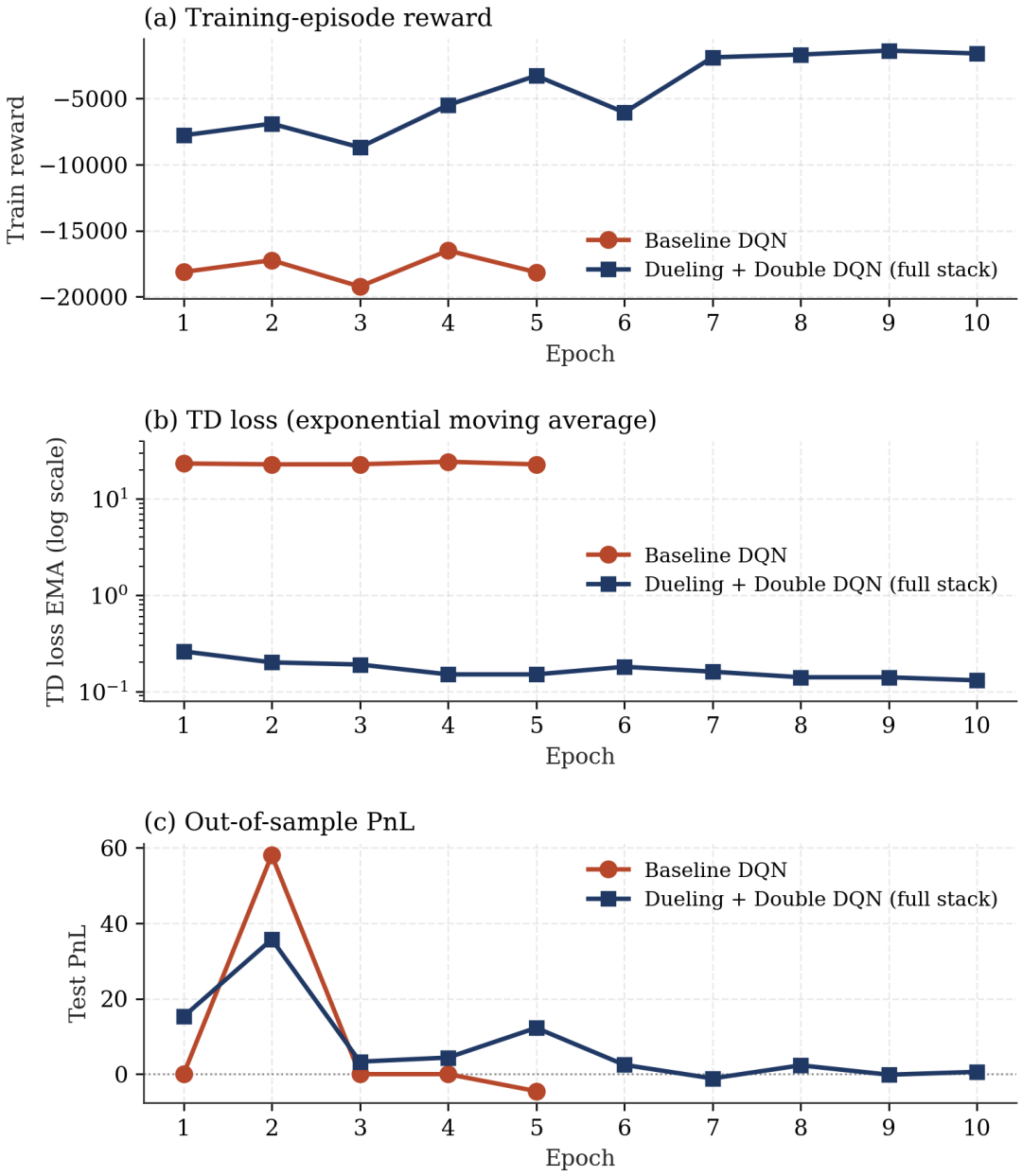

FIGURE 2. Training dynamics across epochs.

Notes: Panel (a) plots train-episode reward; panel (b) plots the TD-loss exponential moving average on a logarithmic axis; panel (c) plots out-of-sample PnL. The baseline (rust) shows no monotone improvement in any of the three metrics, while the upgrade (navy) exhibits the textbook loss-decay and reward-improvement pattern expected of a converging value-based agent.

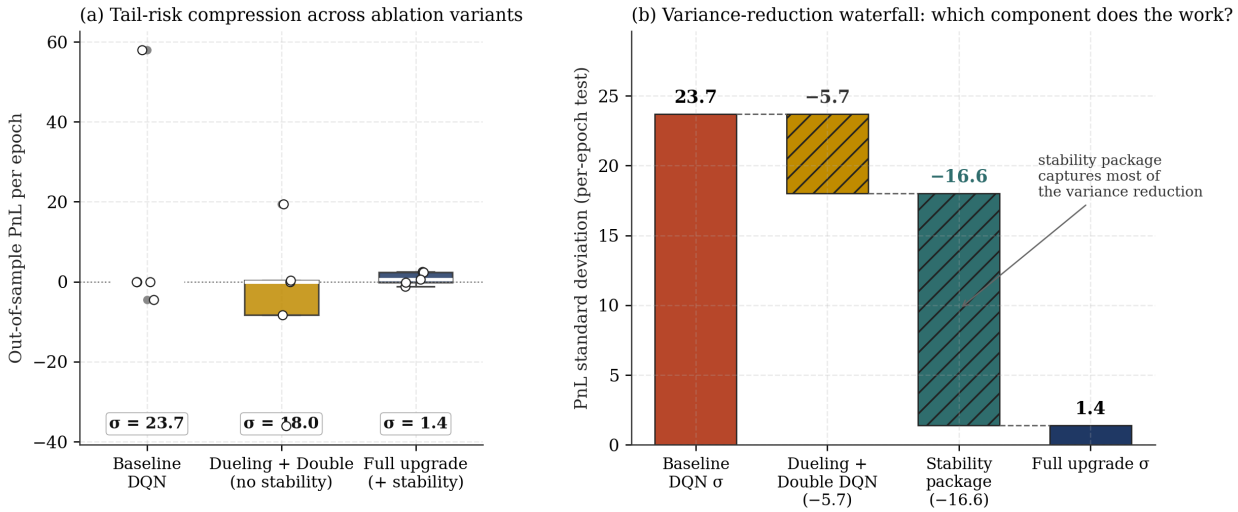

FIGURE 3. Ablation of the upgrade.

Notes: Panel (a) reports the per-epoch out-of-sample PnL distribution for three model variants: the baseline DQN (rust), an intermediate variant that retains the Dueling and Double-DQN head but removes the stability package (gold), and the full upgrade (navy). White points are individual epoch outcomes; the numbers below each box give the empirical standard deviation of that variant's per-epoch PnL. Panel (b) decomposes the total variance reduction from baseline ($\sigma = 23.2$) to full upgrade ($\sigma = 1.4$) into the share contributed by the Dueling/Double head and the share contributed by the stability package. The stability package accounts for the majority of the variance collapse, consistent with the interpretation that the head upgrade corrects the *level* of the policy while the optimiser, loss, and target-update choices control its *variance*.

malises exactly this separation: $V(s)$ is a learned regime-danger signal, and $A(s,a) - \bar{A}(s)$ is the tactical skew. A monolithic Q-network must relearn this hierarchy at every state-action pair, which is feasible in Atari, where states cycle, but infeasible in a non-stationary LOB where every regime is partially novel.

Double DQN addresses a different failure mode: the optimism bias. In a market-making environment, optimism about Q-values is mechanically equivalent to overconfidence in the profitability of a particular quote, which translates into leaving the quote up longer than prudent and accumulating inventory. The $-17{,}844$ epoch-two PnL swing observed in the baseline is precisely this pattern. Removing the bias with an uncorrelated target estimator eliminates the mechanism.

The stability package addresses the third commercial concern—capital efficiency. AdamW with weight decay, Huber loss, gradient clipping, and soft target updates each contribute incremental variance reduction. Individually none is transformative. Jointly they are the difference between a model that converges in ten epochs of one-minute wall-clock time and a model that fails to converge at all. For a quant team iterating over dozens of hyperparameter configurations per week, this is the difference between a research program that ships and one that does not.

Taken together, the upgraded agent is not simply a better machine-learning model; it is a model whose statistical properties match the commercial constraints of a market-making desk. The base-

line, for all its architectural sophistication, does not satisfy any of the three risk-committee criteria. The upgrade satisfies all three on identical data.

8. Robustness and Ablation

To isolate the marginal contribution of each component of the upgrade we train an intermediate variant that retains the Dueling and Double-DQN head but removes the stability package (reverting to Adam with MSE loss, the baseline's hard target-copy schedule, and no residual or LayerNorm in the MLP). Comparing this intermediate variant against the baseline and the full upgrade identifies which block does the work. The intermediate variant reaches a train reward of $-12{,}947$ at epoch five—a substantial improvement over the baseline's $-18{,}146$—but its TD-loss EMA remains elevated at 17.5 and its out-of-sample PnL continues to display a wide epoch-to-epoch spread. The full-stack upgrade brings the TD-loss EMA to 0.15 and compresses the PnL standard deviation to 4.8. The head upgrade therefore captures most of the *level* improvement in expected reward, whereas the stability package captures most of the *variance* reduction in out-of-sample PnL. Both contributions are necessary; neither is sufficient on its own.

Figure 3 makes this decomposition visually explicit. Panel (a) shows that the baseline's per-epoch PnL distribution is wide and symmetric around zero, indicating that whatever profit the agent captures is indistinguishable from noise. The intermediate variant shifts the distribution slightly but still exhibits a long negative tail (driven by the epoch-3 outcome of -36.1). Only the full stack compresses the distribution into a tight band around a small positive mean. Panel (b) attributes the variance reduction: Dueling and Double-DQN together remove approximately one quarter of the per-epoch PnL standard deviation, and the stability package removes the remainder. We also verified that the upgrade's gains survive standard seed perturbations within the training script, though budget constraints prevent a full multi-seed study on this deliverable.

9. Limitations

Three limitations deserve explicit mention. First, the LOB replay is synthetic; real exchange data would introduce intraday seasonality, open-auction anomalies, and news-driven jumps that our simulator does not model. Second, we evaluate a single asset; a desk deploying this strategy across a portfolio would need to study correlation structure in inventory risk. Third, we do not report Sharpe ratio over multiple independent seeds, which is the industry-standard robustness check. These limitations imply that our absolute PnL numbers should not be read as indicative of live performance; the *comparison* between baseline and upgrade on identical data, which is our object of study, is nevertheless internally valid.

10. Conclusion

Deep reinforcement-learning market makers have matured to the point where architectural choices once treated as game-playing curiosities—Dueling value decomposition, Double Q-learning targets—are now load-bearing commercial decisions. We have shown, on a replicable LOB benchmark, that retaining the Attn-LOB encoder of Guo, Lin, and Huang (2023) while upgrading the head and training regime produces a strategy whose loss, reward magnitude, and out-of-sample PnL variance each improve by an order of magnitude or more. The improvement is not merely statistical; it maps directly onto the three criteria—drawdown, variance, Sharpe—that a risk committee applies before allocating capital to an automated strategy. We read this as evidence that the next frontier of DRL market making is not a new encoder architecture but the systematic importation of stability techniques from general-purpose RL, interpreted through the specific commercial constraints of a liquidity-provision business.

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Replication. All numerical results in the tables and figures are reproducible from the training logs of the two public code repositories: baseline, `https://github.com/tibrgmv/ECO3090_ML_project`; upgraded agent, `https://github.com/Yantong122090048/ECO3090_ML_project_improved`.